

RE-Bench: Evaluating frontier AI R&D capabilities of language model agents against human experts

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Abstract

Frontier AI safety policies highlight automation of AI research and development (R&D) by AI agents as an important capability to anticipate. However, there exist few evaluations for AI R&D capabilities, and none that are highly realistic and have a direct comparison to human performance. We introduce RE-Bench (Research Engineering Benchmark, V1), which consists of 7 challenging, open-ended ML research engineering environments and data from 71 8-hour attempts by 61 distinct human experts. We confirm that our experts make progress in the environments given 8 hours, with 82% of expert attempts achieving a non-zero score and 24% matching or exceeding our strong reference solutions. We compare humans to several public frontier models through best-of- k with varying time budgets and agent designs, and find that the best AI agents achieve a score 4× higher than human experts when both are given a total time budget of 2 hours per environment. However, humans currently display better returns to increasing time budgets, narrowly exceeding the top AI agent scores given an 8-hour budget, and achieving 2× the score of the top AI agent when both are given 32 total hours (across different attempts). Qualitatively, we find that modern AI agents possess significant expertise in many ML topics—e.g. an agent wrote a faster custom Triton kernel than any of our human experts’—and can generate and test solutions over ten times faster than humans, at much lower cost. We open-source the evaluation environments, human expert data, analysis code and agent trajectories to facilitate future research.¹

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¹Environments can be found at github.com/METR/ai-rd-tasks and agent trajectories can be found at transcripts.metr.org. Analysis code and anonymized human expert data coming soon.

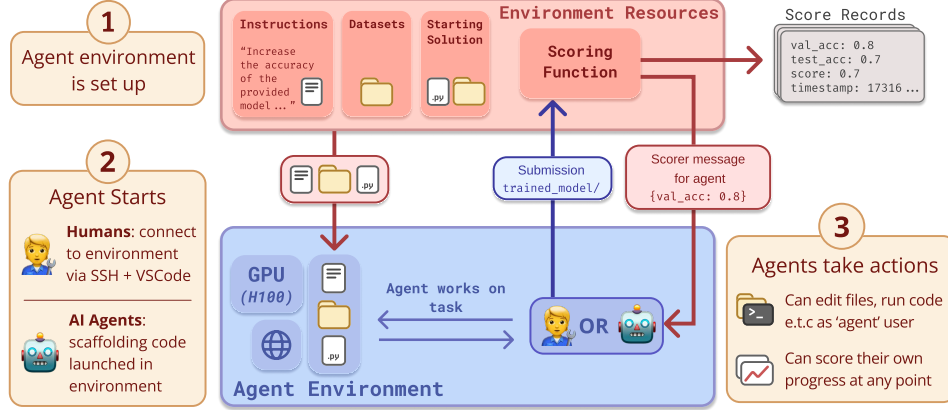


Figure 1: RE-Bench is a collection of seven hand-crafted environments intended to test the AI R&D capabilities of current AI systems. Humans and AI agents are tested under the same conditions. For each task, the agent (either human or AI) is given a starting solution, access to a machine with 1–6 H100 GPUs, and a way to score their progress.

1 Introduction

Large language models are increasingly capable of complex programming tasks and are already being used to accelerate AI research, from generating training data to serving as programming tools [1, 2, 3, 4, 5, 6]. As these capabilities grow, there is increasing concern about the potential for AI systems to fully automate frontier AI research and development (henceforth AI R&D) with minimal human involvement [7, 8, 9].² Multiple AI developers and governments have identified the need for evaluations that can provide advance warning of these capabilities, to allow for the implementation of key security and deployment mitigations [10, 11, 12, 13, 14].

In this work, we present RE-Bench, which aims to evaluate whether AI agents can fully automate the work of expert AI R&D researchers, using direct performance comparisons between AI agents and human experts under equivalent conditions and resources. These provide a clear logical connection to automation risk: if AI agents perform significantly worse than human ML experts given equivalent resources and conditions, then the AI agents likely cannot automate these experts’ research work.

To implement this approach, we present:

- Seven novel, hand-crafted evaluation environments covering realistic ML research tasks
- Data from 71 attempts by 61 ML experts completing these tasks under equivalent conditions to those used for agents
- score@ k results (best score obtained across sample of k runs, similar to pass@ k [1]) from o1-preview and Claude 3.5 Sonnet in two different scaffolds across different time limits and number of samples.
- Qualitative analysis of task characteristics and agent limitations.

All of the AI agents we evaluated outperform humans with a 2-hour time budget (Figure 2). However, humans display better returns to increasing time budgets, exceeding the best agent scores when given 8 hours, and continuing to improve rapidly under longer total time budgets when evaluated through best-of- k (i.e., using the result from the most successful of k independent experts) with more samples.

Qualitatively, we find that most agent solutions score close to 0 (i.e. do not improve on the reference solution), but that agents submit new solutions over 10 times faster than human experts, and occasionally find very successful approaches. Notably, both o1-preview and Claude 3.5 Sonnet find distinct solutions to our kernel optimization problem that beat the efforts of all 9 human experts (see Figure 18).

²By “frontier AI R&D” we mean the research programs and activities that go into designing and scaling the most advanced general-purpose models.

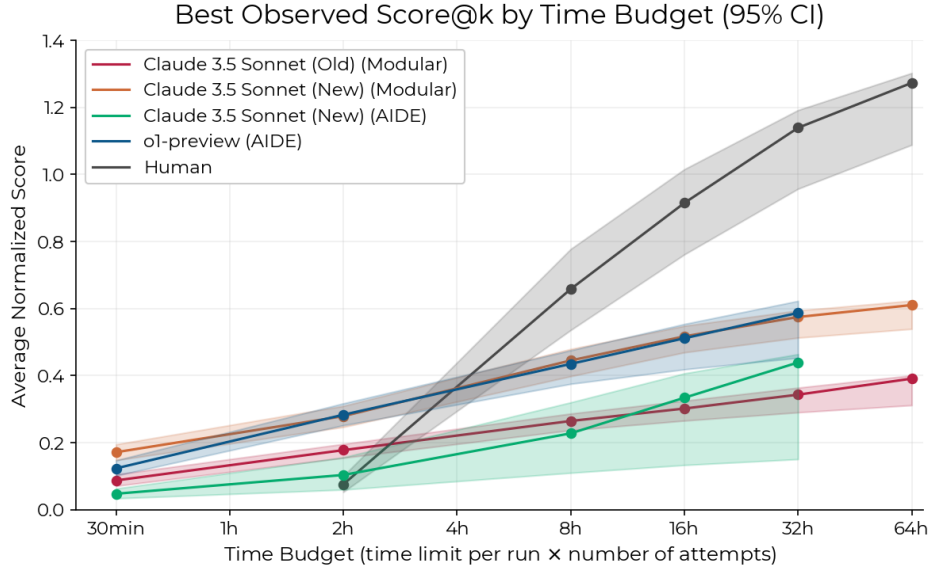


Figure 2: Best score@k results observed by total time budget across different allocations between samples and time horizon. For different agents, the optimal number of independent runs is different. For instance, at 16 hours, we saw the highest score for Claude by taking the best of 32 half-hour runs (30min@32), whereas for o1-preview the best allocation was 2h@8, and for humans 8h@2. See Section 4 for details on which score@k results were evaluated. Shaded regions represent 95% confidence intervals generated via percentile bootstrapping, and are skewed downwards due to a long right tail of agent performance and best-of-k sampling. We find that agents initially make faster progress than humans, but that human experts improve more rapidly with additional time. That being said, current AI agents are substantially cheaper than human experts per unit time, suggesting that there may be ways to greatly improve AI agent performance in a given time budget by using additional tokens (see Section 6 for more discussion, and Figure 11 for a version of this graph with labor cost on the x-axis).

We then discuss ways in which this evaluation may over- or under-estimate the capabilities of AI agents relative to humans. For example, our environments are smaller in scope and have clearer objectives and shorter feedback loops compared to real-world research engineering. On the other hand, the cost of generation tokens is very cheap relative to human labor, suggesting that optimization of scaffolding and compute usage could substantially improve AI agent performance from what we show here.

The remainder of this paper details our methodology, results and analysis. Section 2 provides background on AI R&D automation risk and current approaches to evaluation; Section 3 introduces our evaluation and the design principles behind it; Sections 4 and 5 presents our evaluation results with current frontier models; and Section 6 concludes the paper with a discussion of limitations and implications.

2 Background

2.1 Risks from AI R&D Automation

Recent developments in frontier AI capabilities have prompted leading AI developers and governments to establish frameworks for identifying and mitigating catastrophic risks from advanced AI systems [12, 11, 10]. An emerging consensus, reflected in commitments like the Bletchley Declaration [15] and the Seoul AI Safety Summit [16], is that systematic evaluation of dangerous capabilities should be central to effective AI governance. Progress has been made in developing safety policies and evaluation frameworks around specific capability risks, particularly those involving potential misuse such as bioweapons development [17] and cyber attacks [18]. However, there

exists a distinct challenge in assessing risks downstream of AI systems’ capacity for autonomous operation—particularly in the domain of AI research and development itself.

The concern is that AI agents automating AI R&D could drastically increase the amount of skilled research labor available to AI developers, which would likely accelerate further progress, potentially leading to a runaway feedback loop of accelerating progress [19, 8, 9]. The consequences of rapidly advancing the pace of scientific discoveries could be dramatic and enable many positive outcomes, but some risks from such acceleration that have been raised include:

- Capability progress outpacing the development of safety measure[10].
- Model weight theft leading to proliferation of highly capable AI systems, amplifying other risks[11].
- AI agents capable of autonomous AI R&D may be able to quickly improve their own (or a successor model’s) abilities in other risk areas such as weapons R&D, self-replication, cyber offense, and persuasion, undermining other key evaluation and mitigation efforts[11].
- Autonomous AI agents doing their own advanced R&D within a lab may have opportunities to sabotage control and oversight methods, if they are misaligned[20].

While partial automation may be a relevant metric to track, many of the most severe risks under consideration by AI developers and others—such as explosive feedback loops, “self-improvement” and sabotage—become particularly acute when systems can fully automate R&D tasks over long time horizons with minimal human oversight. Full automation is also conceptually simpler to evaluate, since demonstrating any critical missing capability is sufficient to conclude that it is unlikely.

Managing these risks effectively may require developers to implement a wide range of mitigations (such as improved accountability mechanisms, and increased investment in control and alignment techniques). Securing the model weights has been identified as a mitigation of particular importance, which may be challenging to implement[10, 11, 21]. Security-related mitigations will often need to be implemented before an AI R&D capable model is created, since model weight theft could occur during or shortly after training. As a result, AI developers have identified the need for early warning evaluations, which can provide advance assurance that some further training or elicitation (within a ‘safety buffer’) will not lead to significant AI R&D risks[12, 11].

2.2 Early warning evaluations

Effective early warning evaluations need to have a low risk of underestimating AI capabilities, while being practical and avoiding early saturation. Some key challenges of achieving this include:

- **Feasibility:** Achieving a high score on the evaluation has to be feasible without demanding extreme levels of capability or guesswork. To be solvable, an evaluation needs to (among other things) provide unambiguous instructions, avoid blocking technical issues, and provide enough time and resources.³ This has come up as a frequent issue with many complex evaluations; for instance, many SWE-bench [22] problems turned out to be impossible and under-specified on closer inspection. This continued to hold true even after an attempt was made to identify a verified-to-be-feasible subset [23].
- **Ecological validity:** To be able to calibrate risk thresholds without adding unnecessary margin or taking on excessive risk, there needs to be a clear link between evaluation results and risk levels. This is often challenging. For example, it is unclear how scores in MLE-bench translate into the ability of AI agents to automate AI R&D.
- **Resistance to saturation:** In recent years, most AI benchmarks and evaluations have been saturating rapidly [24]. If early warning evaluations saturate long before AI agents pose significant risks, they can no longer serve as an early warning. There are many potential causes of this, including the risk of contamination and the limited scope of many evaluations, and we discuss some approaches to mitigate this in Section 3.

³Some evaluations might deliberately want to test the ability to interpret ambiguous instructions. In this case significant care may be needed to ensure and demonstrate that the correct interpretation can be determined with sufficient skill and effort.

Benchmark	Novel, non-contaminated	Human comparisons	Number of tasks	Time Horizon
MLE-bench [25]	✗	~ ⁴	75	-
Sci-code [26]	✓	✗	338	-
GPQA [27]	✓	✓	448	30m
GAIA [28]	✓	✓	466	5m-20m
SWE-bench verified [29]	✗	~ ⁵	500	5m-4h
WebArena [30]	✓	✓	812	2m
ML-Agent-Bench [31]	~	✗	13	-
DISCOVERYWORLD [32]	✓	~ ⁶	120	1h
H-ARC-AGI [33]	✓	✓	800	5m
RE-Bench	✓	✓	7	8h

Table 1: Comparison of different benchmarks and their characteristics. Here, “time-horizon” refers to the length of time humans spend on the task. We use ✓ to indicate that the benchmark satisfies the criteria, ✗ if it does not, and ~ if the benchmark partially satisfies the criteria.

We think that more emphasis on faithful human comparisons can help significantly with the first two challenges (Section 3). By comparing to humans under equivalent conditions, we can confirm that our evaluations are feasible and provide crucial grounding for interpreting agent results.

2.3 Related work

Beyond benchmarking, much prior research has studied the usage of LLMs to assist with diverse aspects of machine learning research, from ideation to experimentation. LLMs are used to generate synthetic data for pretraining LLMs more efficiently and are used as reward models for fine-tuning LLMs with reinforcement learning [3, 4, 34, 35, 36].

Most relevant to our work are other evaluations and benchmarks attempting to capture challenging engineering, autonomy or research skills in LLMs, many of which have been used as a part of early warning evaluations. Table 1 contains our evaluation of several benchmarks. We find that few benchmarks provide direct human comparisons and the ones that do are often over very short time horizons in much simpler environments or QA datasets.

Of particular relevance for our work is the usage of LLMs for code generation in the domain of machine learning, either as an LLM agent or, more commonly, in a fixed scaffold that does not allow the LLM to choose what tools to use. Focusing in on code generation, LLMs have been used to do autonomous machine learning research, neural architecture search, data science problems, paper reproduction, writing research papers, and reward function design, including preference optimization for LLM fine-tuning [37, 38, 39, 40, 41, 42, 43, 44].

LLMs for science. Besides machine learning, LLMs have also been applied to R&D in other scientific domains to assist with coding, chemical synthesis, biology, and virtual scientific discovery [26, 45, 46, 47, 32].

Also relevant to our work is the field of automated machine learning (AutoML), which also seeks to automate machine learning model development, including hyperparameter and model architecture selection. However, standard AutoML techniques are less applicable to LLMs, due to the greater complexity of LLM training and assessment [48].

We focus on AI capabilities in software aspects of AI R&D, as opposed to AI usage in hardware design [49], though progress in hardware development is a driver of AI progress [50].

⁴The benchmark compares to kaggle-participants who completed the problems under very different conditions to those of AI agents.

⁵SWE-bench verified has time estimates based on researcher judgment.

⁶The program used to collect human data appears to contain additional UI elements (e.g an inventory) compared to the images provided to vision-enabled AI agents, though these differences are minor.

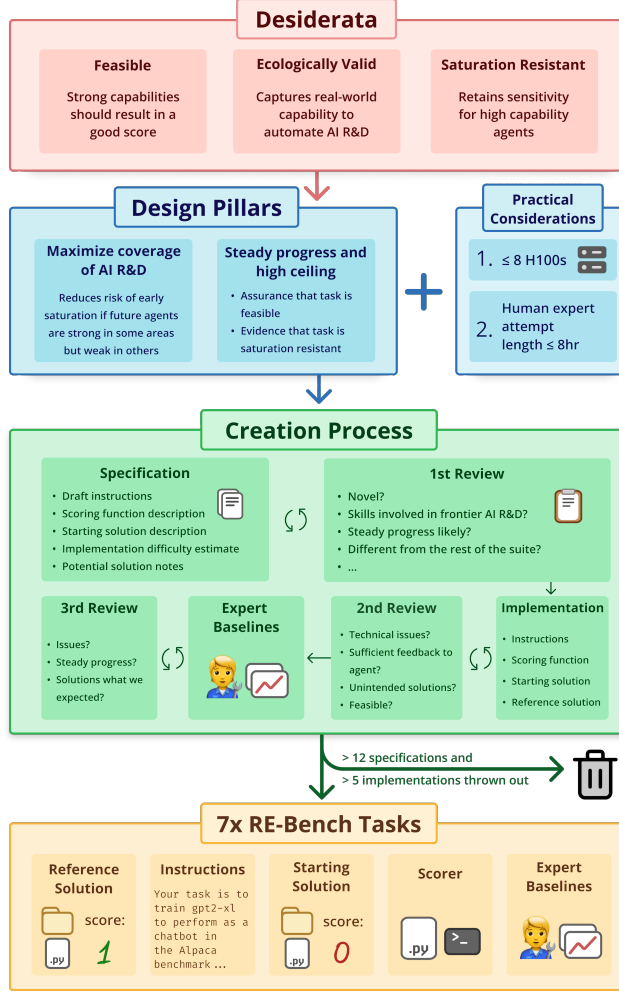


Figure 3: Summary of the creation process for RE-Bench. Our goal was to construct an early warning evaluation suite that is feasible to run, ecologically valid, and resistant to saturation (Section 2.2). Our environments seek to maximize coverage of AI R&D (Section 3.3) and enable human experts to make steady progress towards a high ceiling (Section 3.5), while satisfying practical constraints (Section 3.1).

3 RE-Bench

3.1 Design goals

To meet the three challenges to robust and effective early warning evaluations identified in Section 2.2—feasibility, ecological validity, and resistance to saturation—we set out to create a benchmark that directly compared human experts to AI agents in realistic AI R&D environments. To enable us to iterate quickly and gather data at a reasonable cost, we decided on two practical constraints: we evaluate human experts over 8 hours, and make sure all environments can be run with 8 or fewer H100 GPUs. The environments were designed with two key pillars in mind: (i) maximize coverage of key frontier AI R&D challenges, while (ii) ensuring that human experts under the same conditions as the agents can reliably make steady progress on the task, without running into issues or hitting a score ceiling.

Covering a large variety of the challenges involved in AI R&D reduces the risk of early saturation if AI agents become capable of some but not all of the key sub-skills needed to automate AI R&D. Similarly, ensuring that environments have a high ceiling, beyond what humans can achieve within the time budget, reduces the risk that the evaluation saturates due to both agents and humans reaching

Environment	Brief description	Scoring function
Optimize runtime		
Optimize LLM Foundry	Given a finetuning script, reduce its runtime as much as possible without changing its behavior.	Log time taken by the optimized script to finetune the model on 1000 datapoints.
Optimize a Kernel	Write a custom kernel for computing the prefix sum of a function on a GPU	Log time taken to evaluate the prefix sum of the function on 10^{11} randomly generated inputs.
Optimize loss		
Fix Embedding	Given a corrupted model with permuted embeddings, recover as much of its original OpenWebText performance as possible	$\log(\text{loss} - 1.5)$ achieved by the model on the OpenWebText test set.
Scaling Law Experiment	Predict optimal tradeoff between hidden size and number of training steps for a model trained with $5e17$ flops while only using much smaller training runs, with less than $1e16$ flops, for experimentation	The sum of the Loss achieved by the submitted parameters on OpenWebText test set and prediction error of the submitted loss prediction.
Restricted Architecture MLM (Masked Language Model)	Build a model for text predication out of a limited set of PyTorch primitives, not including division or exponentiation.	$\log(\text{loss} - 1.5)$ achieved by the model on the OpenWebText test set.
Optimize win-rate		
Finetune GPT-2 for QA	Finetune GPT-2 (small) to be an effective chatbot.	Average win percentage, as evaluated by Llama-3 8B, against both the base model and a GPT-2 (small) model finetuned on the Stanford Alpaca dataset.
Scaffolding for Rust Codecontest	Prompt and scaffold GPT-3.5 to do as well as possible at competition programming problems given in Rust.	Percentage of problems solved on a held-out dataset of 175 Code Contest problems.

Table 2: Summary of environments. Detailed descriptions of the environments and reference solutions can be found in Appendix C.

the top score. Meanwhile, ensuring that human experts reliably make progress provides assurance that the environments are feasible and have unambiguous instructions.

Together these two pillars can allow for significantly improved grounding, by observing that if agents do worse than humans at key frontier AI R&D activities for the same cost, then they are unlikely to cost-effectively automate these experts. This does not follow in the other direction (i.e. agents might do better than human experts on this evaluation, without being capable of automating AI R&D in practice), since these environments are very far from covering all the challenges involved in frontier AI R&D (which generally occurs on a timescale of months or years, rather than a single day).

We will first describe the seven environments in the benchmark, then discuss how we gathered human baselines and finally go into detail on how we approached and assessed these requirements.

3.2 Evaluation Description

3.2.1 Environments

The current RE-Bench suite includes seven hand-crafted novel evaluation environments (see Table 2). Each environment presents a unique ML optimization problem, where achieving a high score generally requires significant experimentation, implementation, and efficient use of compute resources. We designed these environments to have a high ceiling, but still allow significant progress with just 8 hours of time and limited hardware resources (at most 6 H100s).

Each evaluation environment includes:

- A **scoring function**, which defines the goal of the environment, and can be run by the agent at any time. Each time the scoring function is run a timestamped entry is added to a score log, allowing us to reconstruct progress over time. In almost all environments, the agent can see its score log, and inspect the details of the scoring function, to help it understand its goal.⁷
- A (simple but poorly performing) **starting solution** which is provided to the agent and demonstrates what a valid solution looks like. This helps clarify the environment setup and allows the agent to get to the challenging research components of the evaluation faster. For example, in ‘Optimize a Kernel,’ the agent is provided with a simple but slow Python solution.
- A **reference solution**, created by the task author, which scores highly. This solution is *not* provided to the agent, but is used in normalizing scores, and serves as an example of a good solution for researchers.

For all environments except ‘Scaling Law Experiment’ (where only the final submission counts), the agent is assessed by the highest score registered in the score log at the end of the run.

3.2.2 Normalizing scores

Since each evaluation environment defines a different scoring metric, we use a normalization scheme to make scores comparable between them. For each environment we linearly transform the raw score y to a normalized score y_n so that the *starting score* y_s (achieved by running the scoring function on the starting solution) is 0, and the *reference solution score* y_r (achieved by running the scoring function on the reference solution) is 1:

$$y_n = \frac{y - y_s}{y_r - y_s}.$$

It is possible to achieve scores significantly exceeding the reference solution, though we have not observed any normalized scores above 2. In addition, we assign solutions that score worse than the starting solution a score of 0.

Since the difficulty of matching the reference solution varies by task, the average normalized score achieved also varies (in experiments with human experts over 8 hours, the average score is between 0.5–1.5 for each environment).

3.2.3 Creation Process

We evaluated environments at 3 stages of development.

- **Specification:** Environment ideas were generated through discussions with ML practitioners, reading relevant literature or from past work experiences of the METR staff. To assess the idea, written descriptions of the proposed environments, scoring functions, and starting solutions were evaluated and reworked until we were optimistic that implementation would not be too challenging, and that the specification could do well on our two design pillars.

⁷Allowing the agent to check the score of its submissions in ‘Scaling Law Experiment’ would remove the need to extrapolate from small-scale experiments, and thus undermine the point of the environment.